

Put the Council on Trial

A falsifiable test of roles, robots and topology

Same model. Frozen artifacts. Hidden defects. Blinded scoring.



Three claims — not one

1 · Roles

Independent specialists find more real defects than repeated general reviews.

2 · Personas

Character wrappers add value beyond the explicit specialist checklist.

3 · Topology

Fan-out + one fusion retains more signal than serial hand-offs.

Black-box limit: we measure repeatable shifts in defect coverage, errors, variance and cost. Role cues may change hidden-state and attention trajectories, and which learned features contribute through fixed weights; this changes logits and the conditional token distribution; decoding then adds sample variance. API outputs cannot isolate those stages or prove that different weights were “activated.”

Three synthetic traps

Revenue dashboard

01

SQL · dbt · Cube · recovery

Fan-out, full scans, PII, contract bypass, speculative factory, forgotten flag, vague safety.

Key rotation at 03:00

02

runbook · deployment note

Leaked key, outage window, wrong context, latest, muted alerts, no rollback, four-service factory.

Webhook redesign

03

architecture proposal

Race-prone dedupe, slow ACK, payload logging, six queues, schema drift, silent backlog, lossy pause.

Packet = one frozen review scenario and its context · 8 planted defects · private answer key

First: one reviewer

S0

Plain generic review

Weak baseline

S1

Functional omnibus review

All seven lenses listed fairly · no persona

S2

Functional omnibus + GLaDOS

Length-matched character layer

Single-call persona effect

S2 - S1

Fair baseline

S1, never S0 alone

Arm = one prompt and orchestration condition tested against every packet

Then match the calls

M0

Repeated functional omnibus

7 independent copies of S1

Controls for more samples

M1

Functional specialists

7 explicit lenses, no fiction

Tests role specialisation

M2

Robot specialists

Same 7 kernels + character wrappers

Tests pop-culture increment

seven fresh review calls

→

same model · same fuser

→

ranked verdict

Every arm uses one pinned cheap model first. Same artifact, schema, caps and fresh sessions.

Each subtraction answers one question

M1 – M0

Specialisation beyond repeated general sampling

M2 – M1

Pop-culture wrapper beyond the same role instructions

M0 – S1

More samples without specialist diversity

M2 – S1

Practical council uplift — with cost reported

KEY

S1

Functional omnibus

1 call

M0

Repeated functional omnibus

7 reviewers + fuser

M1

Functional specialists

7 reviewers + fuser

M2

Robot specialists

7 reviewers + fuser

If M2 only produces more colourful prose, it loses. Count additional **true defect IDs**, not personality.

Score signal, noise and cost

Primary

Macro F1 on planted defects

Diversity

Jaccard overlap · unique true findings · category coverage

Cone claim

p10 · worst case · variance · IQR

Efficiency

true findings / 1k tokens · \$ · latency

A unique false positive is noise. Duplicate true findings are consensus. Different wording is neither.

Did the arbiter help — or drop the minority report?

Raw union

All true and false findings from seven reviewers

maximum coverage

maximum noise

Fused verdict

Same fuser de-duplicates, ranks and rules

precision gain?

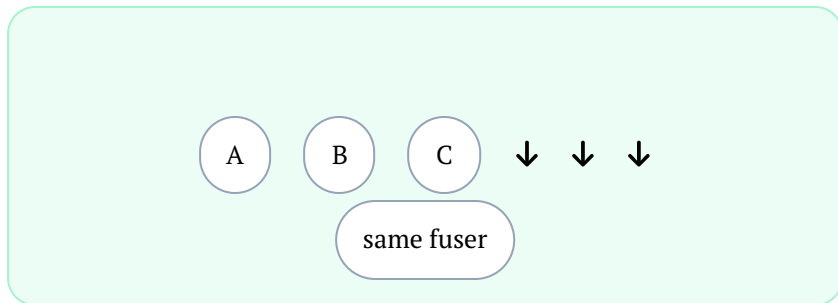
fusion loss?

Fusion retention = fused true positives $\div$ raw-union true positives

Score both outputs. A neat summary that drops valid edge cases weakens the arbiter claim.

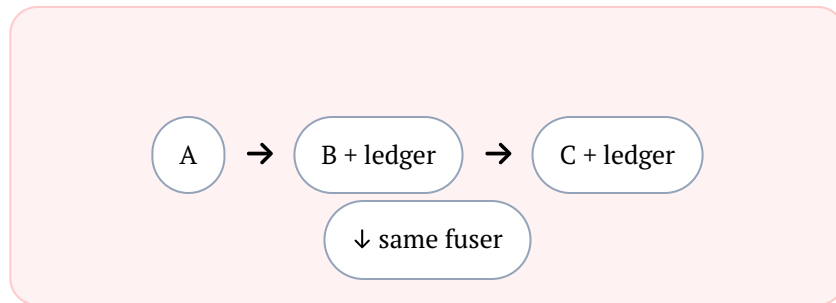
Fan-out versus every chain order

FAN-OUT



Each reviewer sees the original packet independently.

INFORMED CHAIN



Every reviewer still sees the original. Prior findings can anchor, help, or vanish.

3 roles → 6 orders

ABC · ACB · BAC · BCA · CAB · CBA

Measure true findings added, retained and dropped at each hop — plus final F1 and order sensitivity.

Then cross the provider boundary



Send the **byte-identical K-2SO prompt** and its persona-free twin to each pinned model. Test `provider × persona` before paying to replicate the whole council.

Same text ≠ same tokens or hidden system prompt. Provider is a factor, not a verdict.

Let the robots lose.

Packet one frozen review scenario

Arm one prompt/orchestration condition: S0–M2

Smoke: run every arm once on every packet; repair prompts and ceiling/floor effects.

Pilot: 10 samples per packet $\times$ arm; 3 samples per topology order.

Confirm: freeze packet variants and thresholds; score blind against private answer keys.

Roles win
 $M1 > M0$

Characters earn rent
 $M2 > M1$ per token

Or we delete the bit
nulls count

`experiment/protocol.md` · frozen scenarios under `experiment/scenarios/`